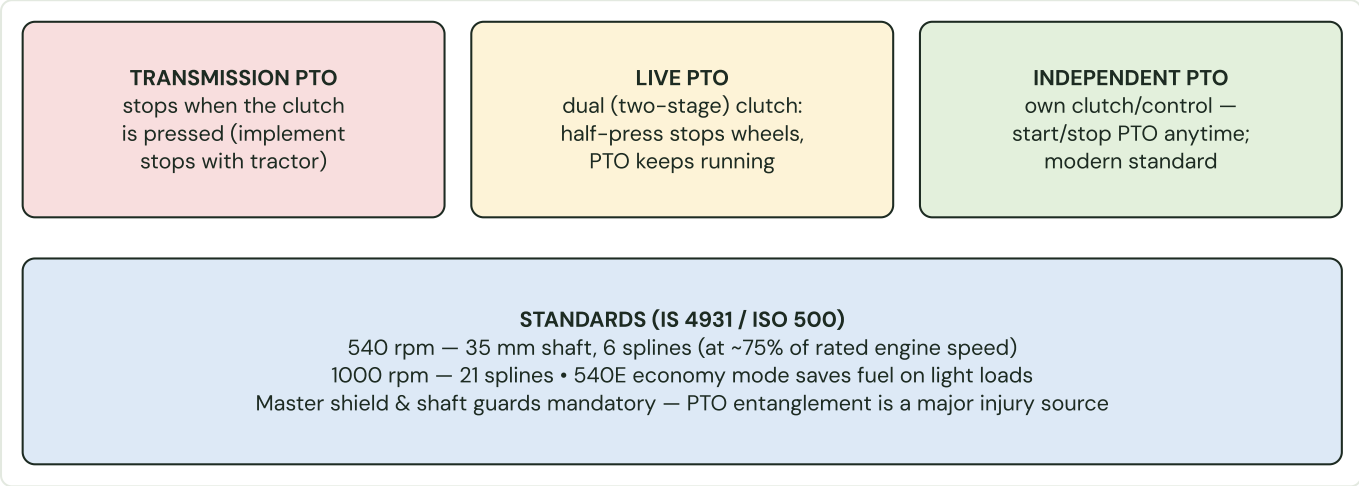
UNIT 4 · TOPIC 6

PTO, Drawbar and Belt Pulley

Syllabus topic: Power outlets: P.T.O. and drawbar

Power take-off (PTO)

The PTO drives rotavators, threshers (via shaft or pulley), balers, sprayers and mowers.



PTO types and standards (redrawn from the source figure)

Standard speed	Shaft (IS 4931 / ISO 500)
540 rpm	35 mm, 6-spline
1000 rpm	21-spline
540E (economy)	540 rpm at a lower engine rpm, for light loads

- Standard PTO speed is reached at ~**75%** of rated engine speed.

PTO type	Behaviour
Transmission	Pressing the clutch stops BOTH wheels and PTO
LIVE	Dual clutch — ground drive and PTO released in two stages
INDEPENDENT	Own clutch — engage/disengage any time
Ground-speed	Rotation proportional to forward travel — for planters

- PTO power ≈ **87–92%** of engine power.
- **Safety:** master shield and tunnel guard on the shaft; never step over a turning shaft (Unit 5).

Drawbar

- **Swinging drawbar** (pivots, lateral holes) for trailed implements — it aligns with the line of pull on turns; or a fixed under-slung bar.
- Height ~**330–500 mm**, adjustable. Hitch height sets weight transfer and stability (Unit 2, Topic 4; Unit 6; Topic 9a here).
- Drawbar power = tractive output after wheel-slip losses, measured in drawbar tests (Topic 11).

- The three-point linkage (Cat I/II per IS 4468/ISO 730) carries mounted implements under hydraulic control (Topics 9b and 10).

Belt pulley

The historic outlet for chaff cutters and threshers by flat belt — largely replaced by the PTO. Standard belt speed is in the ~3100 ft/min class (a fact occasionally asked).

Practice MCQs

Q132. Standard PTO speeds are:

(a) 540 and 1000 rpm
(b) 750 and 1500
(c) 300 and 600
(d) 1000 and 2000

Q133. The 540-rpm PTO shaft has:

(a) 21 splines, 45 mm
(b) 6 splines, 35 mm
(c) 8 splines, 25 mm
(d) Keyed 40 mm

Q134. The 1000-rpm PTO shaft has:

(a) 6 splines
(b) 10 splines
(c) 21 splines
(d) No splines

Q135. A PTO that can be engaged/disengaged anytime via its own clutch is:

(a) Transmission type
(b) Live
(c) Independent
(d) Ground-speed

Q136. With a TRANSMISSION-type PTO, pressing the clutch:

(a) Stops only wheels
(b) Stops both wheels and PTO
(c) Stops only PTO
(d) Nothing

Q137. Ground-speed PTO rotation is proportional to:

(a) Engine speed
(b) Forward travel of the tractor
(c) Fan speed
(d) Oil pressure

Q138. PTO power as a share of engine power is roughly:

(a) 50–60%
(b) 70–75%
(c) 87–92%
(d) 100%

Q139. Tractor drawbar height is typically:

(a) 100–150 mm
(b) 330–500 mm
(c) 700–900 mm
(d) 1200 mm

Q140. A swinging drawbar is used because it:

(a) Adds power
(b) Aligns with the line of pull of trailed implements on turns
(c) Cools the diff
(d) Raises speed

Answers

Q132 — (a) 540 and 1000 rpm

Q133 — (b) 6 splines, 35 mm — *IS 4931/ISO 500 — 540 rpm: 35 mm, 6-spline; reached near 75% of rated engine speed.*

Q134 — (c) 21 splines — *IS 4931/ISO 500 — 1000 rpm: 21-spline.*

Q135 — (c) Independent

Q136 — (b) Stops both wheels and PTO

Q137 — (b) Forward travel of the tractor

Q138 — (c) 87–92%

Q139 — (b) 330–500 mm

Q140 — (b) Aligns with the line of pull of trailed implements on turns

Sources: Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; IS 4931/ISO 500 (PTO), IS 4468/ISO 730 (linkage); NPTEL farm power lectures.
